Bin Hu

Networked Systems & Machine Learning | Ph.D. Student, USC

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EDUCATION

University of Southern California (USC)

Ph.D. in Electrical and Computer Engineering | Aug. 2023 - May 2028 (Expected)

Advisor: Prof. Feng Qian. Expected graduation: 05/2028.

Prior doctoral studies: University of Minnesota, CSE, Aug. 2022-Aug. 2023; transferred with advisor.

Southeast University (SEU)

M.Sc. in Computer Science and Engineering | Sept. 2019 - Jun. 2022

Southeast University (SEU)

B.Eng. in Computer Science and Engineering | Sept. 2015 - Jun. 2019

TECHNICAL SKILLS

Programming: Python, C/C++, Java, MATLAB.

Tools & platforms: PyTorch, Linux, WebRTC, Android, Git, Docker, Wireshark/tcpdump, TCP/UDP sockets, Raspberry Pi, GNURadio, SDR.

RESEARCH & ENGINEERING EXPERIENCE

University of Southern California / University of Minnesota

Ph.D. Student | Aug. 2022 - Present

- Satellite network measurement, prediction, and optimization (HotMobile 2026): Co-authored a hierarchical framework combining satellite geometry and network telemetry to predict upcoming connections, connection-window performance, and per-second link dynamics for proactive satellite-aware optimization.
- Collected and synchronized network measurements and satellite information, preprocessed training data, and built PyTorch models for network and satellite prediction.
- Real-system evaluation on Starlink achieved throughput and latency mean absolute errors of 8.37 Mbps and 14.25 ms, respectively, versus 29.54 Mbps and 26.31 ms for a self-similar baseline.
- Satellite-cellular integration (CoNEXT Companion 2023, co-first author): Compared two Starlink configurations and three 4G/5G cellular carriers over 3,800+ km (1,239 tests). The study's buffer-tuned MPTCP emulation achieved 30-66% higher throughput than the better single path.
- Multipath networking: Built client, server, and proxy components using Raspberry Pi relay nodes; increased throughput by 50-80% and reduced packet-loss rate to about one-tenth of a single satellite connection.
- WebRTC video-call optimization (INFOCOM 2026 Poster): Developed a WebRTC-based satellite video-call quality measurement system, using Docker, to support the poster research. Co-authored a prediction-driven design that adapts video transmission and reconstruction to anticipated satellite-link degradation.
- NIER - neural-enhanced low-bitrate video conferencing: The system reconstructs video from sparse key-points and reference images. Contributed to rebuilding its neural model for Android phones.
- Net-JEPA: Contribute to research on a 4M-parameter self-supervised world model that learns reusable network dynamics from telemetry for adaptive bitrate streaming (ABR) and congestion control (CC).
- In trace-driven evaluation, the frozen ABR-pretrained model transfers to CC by fitting only the readout on 20 traces, achieving capacity forecast MSLE of 0.84 versus 1.93 for random initialization in an open-loop fluid-queue emulator.

RESEARCH EXPERIENCE (CONTINUED)

Singapore University of Technology and Design (SUTD)

Research Assistant | May 2021 - Apr. 2022

- Worked on Explainable AI (XAI) for Automatic Modulation Classification (AMC)
- Combined XAI techniques with wireless communication systems
- Designed interpretable ML models for signal classification

Singapore University of Technology and Design (SUTD)

Research Assistant | Jul. 2021

- Designed and implemented an AI-based software modulator
- Achieved low-cost and high-flexibility design

Southeast University (SEU)

Graduate Student Researcher | Sept. 2019 - Oct. 2020

- Studied concurrent communication in LoRa networks
- Designed and evaluated a multi-dimensional demodulator
- Published ICNP 2020 paper on collided LoRa transmission decoding

Southeast University (SEU)

Graduate Student Researcher | Jun. 2021

- Focused on collision detection in wireless transmission
- Implemented fine-grained self-correlation detector
- Published WASA 2021 paper

SELECTED PUBLICATIONS

Predicting Connectivity and Link Performance in Dynamic LEO Satellite Networks. **ACM HotMobile, 2026.** Bin Hu et al.

Prediction-Driven QoE Optimization for Video Calls over LEO Satellite Networks. **IEEE INFOCOM Poster, 2026.** Bin Hu et al.

NN-Defined Modulator: Reconfigurable and Portable Software Modulator on IoT Gateways. **USENIX NSDI, 2024.** Jiazhao Wang et al. (co-author: Bin Hu)

LEO Satellite vs. Cellular Networks: Exploring the Potential for Synergistic Integration. **ACM CoNEXT Companion, 2023.** Bin Hu et al.

SCLoRa: Leveraging Multi-Dimensionality in Decoding Collided LoRa Transmissions. **IEEE ICNP, 2020.** Bin Hu et al.

INDUSTRY EXPERIENCE

Guangzhou Desheng Artificial Intelligence Technology Co. Ltd.

Algorithm Engineer (Intern) | Aug. 2018

- Developed camera-based assisted driving systems
- Implemented driver behavior detection and warning algorithms

SELECTED HONORS

Student Travel Grant, ACM HotMobile 2026; Second Prize, Academic Scholarship, Southeast University (2019, 2020); National Second Prize, National College Student Smart Car Competition (2018)